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LAB 4 – OOP (Java - NetBeans)

Task 1: Addition Class

```
class Addition {
    int num1, num2, res;

    Addition() {
        num1 = 0; num2 = 0; res = 0;
    }

    void setNum1(int n1) { num1 = n1; }
    void setNum2(int n2) { num2 = n2; }

    int getNum1() { return num1; }
    int getNum2() { return num2; }
    int getRes() { return res; }

    int add() {
        res = num1 + num2;
        return res;
    }
}

public class Task1 {
    public static void main(String[] args) {
        Addition A1 = new Addition();
        A1.setNum1(5);
        A1.setNum2(6);
        Addition A2 = new Addition();
        System.out.println("A1: num1=" + A1.getNum1() + ", num2=" +
A1.getNum2() + ", res=" + A1.add());
        System.out.println("A2: num1=" + A2.getNum1() + ", num2=" +
A2.getNum2() + ", res=" + A2.add());
    }
}
```

Task 2: Person Class

```
class Person {
    private String FirstName, LastName, FullName;

    Person() {
        FirstName = ""; LastName = ""; FullName = "";
    }

    void setFirstName(String fname) { FirstName = fname; }
    void setLastName(String lname) { LastName = lname; }

    String getFirstName() { return FirstName; }
    String getLastName() { return LastName; }
}
```

```

        String getFullName()    { return FullName; }

        void ConcatenateName() {
            FullName = FirstName + " " + LastName;
            System.out.println("Full Name: " + FullName);
        }
    }

    public class Task2 {
        public static void main(String[] args) {
            Person P1 = new Person();
            P1.setFirstName("John");
            P1.setLastName("Doe");
            Person P2 = new Person();
            System.out.print("P1 -> "); P1.ConcatenateName();
            System.out.print("P2 -> "); P2.ConcatenateName();
        }
    }
}

```

Task 3: Book Class

```

class Book {
    private String title, author;
    private int publicationYear;
    private int pageCount;

    Book() {
        title = "Unknown"; author = "Unknown";
        publicationYear = 0; pageCount = 0;
    }

    Book(String t, String a, int year, int pages) {
        title = t; author = a;
        publicationYear = year; pageCount = pages;
    }

    void setTitle(String t)           { title = t; }
    void setAuthor(String a)          { author = a; }
    void setPublicationYear(int year) { publicationYear = year; }
    void setPageCount(int pages)      { pageCount = pages; }

    String getTitle()                 { return title; }
    String getAuthor()                 { return author; }
    int    getPublicationYear() { return publicationYear; }
    int    getPageCount()           { return pageCount; }

    void DisplayBookInfo() {
        System.out.println("Title:           " + title);
        System.out.println("Author:          " + author);
        System.out.println("Publication Year: " + publicationYear);
        System.out.println("Page Count:      " + pageCount);
    }
}

```

```
public class Task3 {  
    public static void main(String[] args) {  
        Book book1 = new Book("To Kill a Mockingbird", "Harper Lee", 1960,  
281);  
        book1.DisplayBookInfo();  
  
        Book book2 = new Book();  
        book2.setTitle("The Great Gatsby");  
        book2.setAuthor("F. Scott Fitzgerald");  
        book2.setPublicationYear(1925);  
        book2.DisplayBookInfo();  
    }  
}
```